Knowing All You Need: An Intent-Driven Proactive Assistant for All-Day Wearable Intelligence

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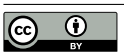
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Abstract

Body-worn cameras extend computing from short observations toward all-day first-person experience. Such observations become useful wearable intelligence only when interpreted with a person's characteristics, life history, and current needs. We present Knowing All You Need, a wearable proactive assistant that provides services from user profiles and life memory. A local profile router selects scene-relevant attributes; lifecycle-aware memory creates, modifies, completes, or cancels temporary tasks. Hierarchical life memory, change-aware triggering, staged retrieval, and evidence-constrained decisions recover facts and choose between response and silence. We conduct three progressive studies on seven days of daily recordings: Study 1 evaluates history-grounded response over all days, while Studies 2 and 3 evaluate profile intents and dynamic tasks on Day 1. The method improves F1 by 0.0502 over the strongest Study 1 baseline. For P01 and P02, segment-level F1 is 7.69%, with response-window recall of 28.57% and 14.29%; dynamic tasks reach 19.35% segment–task F1 and 30.00% window recall. Results support heterogeneous intent memory while identifying intervention timing as the main challenge.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Computer vision tasks.*



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Keywords

wearable intelligence, proactive assistant, intent-driven interaction, long-term memory, ubiquitous computing

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1 Problem Formulation

Wearable cameras move computation into everyday activity, observing objects, places, actions, and how earlier events change the present. An item placed in a drawer should affect later search; a concise-guidance preference may warrant a brief connection check; a refrigeration request made while shopping may trigger only when the refrigerator appears. Wearable intelligence must therefore connect personal characteristics, ongoing needs, and objective history, and intervene only while help remains actionable. We call this **intent-driven proactive assistance**: the system uses intent and evidence to choose what to say, when to say it, or whether to remain silent. This requires jointly handling diverse user intents and long egocentric context.

Diverse user intents. Intent is a user-specific condition governing whom to help, under which constraints, and toward what action. **Profile-driven proactive intent** comes from a user-confirmed profile of personality, behavior, interests, preferences, goals, interaction style, and accessibility needs. These attributes remain relatively stable, but only a few may matter in a scene; the system must derive useful assistance without forcing weak personalization. **Dynamic interaction-task intent** arises when a user creates, modifies, or cancels a temporary request. The system must preserve lifecycle

order, identify active tasks, and trigger once when a requested condition occurs. Profile intent determines useful content, whereas task intent determines state and timing.

Long egocentric context. First-person video is redundant, occluded, and filled with recurring objects and actions. Memory must represent events—object relocation, action completion, plan creation, and interaction—and their effects on future state. New locations supersede old ones; completed reminders must not repeat; canceled requests suppress later responses. Frame matching may confuse preparation with the action point, while poor compression retrieves obsolete state. Long-term memory therefore needs selective writing, state updates, traceable retrieval, and causal online decisions.

Knowing All You Need addresses diverse intents with heterogeneous intent memories and long context with hierarchical life memory, then unifies them through triggering, retrieval, and evidence gating. Its objective is not to maximize how often the assistant speaks, but to increase the net value of each intervention when the user, scene, history, and timing jointly support it.

2 Research Motivation

2.1 Related Work

Proactive agents identify opportunities to help and choose interruption times without a current-turn request. Video captures objects, actions, and environmental change. Here, **intent understanding** means inferring an assistance goal from stable characteristics or temporary constraints, current evidence, and historical state—not classifying a specified question. A response must determine what is needed, when its visual condition holds, and whether the event has changed or completed. A low-sugar preference may justify guidance when desserts become actionable; a refrigerator reminder should fire only while active and when that scene arrives.

Prior work has progressively increased the complexity of interaction in video-based settings. StreamingBench and OVO-Bench pose single-turn questions at different moments to evaluate current understanding, historical recall, or delayed answering, but the question still specifies the response objective in advance [7, 8]. IP-IBench extends interaction to multi-turn requests and proactive task management over continuous processes, while the principal intents remain explicitly supplied through interaction [6]. EgoPro-Bench introduces personalized profiles and requires models to reason about response content and timing from a single user intent [10]. In parallel, Ego4D, HoloAssist, and EgoLife provide large-scale, collaborative, or multi-day egocentric life data [2, 11, 13]; external-memory systems show how persistent memory can extend model information retention [9, 12, 14]. However, a practical wearable assistant must simultaneously handle multiple profile attributes, parallel temporary tasks, and life events that alter subsequent meaning. Existing work has not yet unified such multi-source, multi-timescale, mutable intent sets.

2.2 Motivation

Wearables place past events, present scenes, and future actions on one timeline. Yet incorrect interventions consume attention, and even useful notifications are better at task breakpoints [3, 5]. We seek an intent-driven framework that selects a relevant profile attribute, maintains temporary tasks, retrieves decision evidence from all-day video, and treats silence as an output. Three studies begin with life history, add profile intents, and then dynamic tasks, exposing how failures change with intent complexity.

3 Task Definition

Let the input segment at time t be $x_t = (v_t, s_t, \tau_t)$, denoting the video observation, subtitle, and timestamp. Strict online history is

$$\mathcal{H}_t = \{x_i \mid i \le t\}, \qquad x_{t+1:T} \notin \mathcal{H}_t. \tag{1}$$

Diverse intents comprise stable profile intents and dynamic task intents:

$$\mathcal{I}_t = \mathcal{I}^{pro} \cup \mathcal{I}_t^{dyn}. \tag{2}$$

A profile attribute $p_j = (k_j, v_j, src_j, q_j)$ encodes its type, content, source, and confidence. The type k_j covers personality, interest, preference, goal, or interaction style. A dynamic task

$$d_i = (u_i, c_i, a_i, \tau_i^d, s_i, r_i, q_i) \tag{3}$$

contains the original interaction, trigger condition, expected action, validity period, lifecycle state, reference chain, and confidence, with $s_i \in \{active, completed, canceled, expired\}$. The system selects between candidate responses C_t and silence:

$$a_t^* = \arg\max_{a \in C_t \cup \{\varnothing\}} U(a \mid x_t, \mathcal{H}_t, \mathcal{I}_t) - \lambda C_{int}(a), \tag{4}$$

where U measures evidential support, personalized relevance, and action value, while C_{int} represents the cost of an incorrect or premature interruption.

Under this shared decision objective, three progressive studies control which intent sources the system may access.

Study 1: history-driven basic proactive response. Using only x_t and the available life history $\mathcal{H}_t$, the system determines whether an earlier event makes the present scene actionable. A positive response must be historically supported, timely, and non-duplicate; otherwise the output is $\varnothing$. This study tests long-term retrieval and low-interruption decision making.

Study 2: stable-profile-driven personalized response. We add $\mathcal{I}^{pro}$ and require the system to select a scene-relevant attribute from a complete profile and derive appropriate assistance. It should remain silent under weak relevance, premature timing, or insufficient benefit. This evaluates profile routing, personalized content, and suppression.

Study 3: lifecycle-aware dynamic-task response. We further add $\mathcal{I}_t^{dyn}$. Users may create, modify, or cancel requests; the system resolves references, maintains each current task state, and admits only satisfied *active* tasks. A first valid response marks the task *completed*. This evaluates lifecycle maintenance, timing, cancellation, and one-shot deduplication.

4 Knowing All You Need Framework

Figure 1 summarizes Knowing All You Need. Wearable observations populate life, profile-intent, and dynamic-task memories; change-aware triggering and hierarchical retrieval gather evidence; a constrained decision then responds or remains silent. Outcomes are written back to memory and lifecycle state, forming an all-day closed loop.

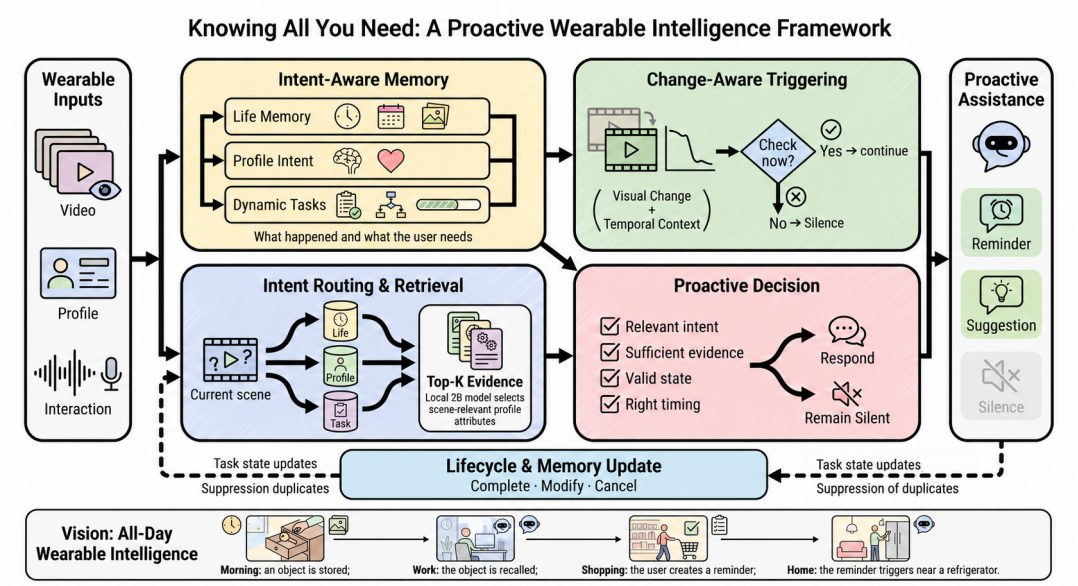


Figure 1: Knowing All You Need framework and all-day wearable vision. Heterogeneous inputs populate intent-aware memories; triggering, retrieval, and evidence-constrained decisions produce assistance or silence, while feedback updates memory and task state.

4.1 Intent-Conditioned Heterogeneous Memory

The perception module converts a segment into

$$\xi_t = f_\theta(v_t, s_t) = (d_t, O_t, E_t, P_t, D_t), \tag{5}$$

where d_t summarizes the scene and O_t, E_t, P_t, D_t denote entities, events, plans, and interactions. Life memory retains these records and their vector indices, replacing entity state while preserving traceable history. A local Qwen3-VL-2B performs profile routing from the current observation, eligible memories, and recent responses:

$$R_t^{pro} = g_{\theta_{2B}}(\xi_t, M^{pro}, H_t^{resp}), \qquad R_t^{pro} \subseteq M^{pro},\ |R_t^{pro}| \le 2. \tag{6}$$

The model returns at most two relevant `memory_ids`; without a candidate, the profile branch remains silent. Dynamic tasks are updated as $M_t^{dyn} = Update(M_{t-1}^{dyn}, y_t, f_t)$ from interaction y_t and feedback f_t. The eligible set $A_t = \{d_i \mid s_i = active, t \in \tau_i^d, c_i(\xi_t) = 1\}$ excludes completed, canceled, and expired tasks, separating life history, enduring user needs, and temporary requirements.

4.2 Proactive Triggering and Hierarchical Retrieval

Rather than invoking a large model for every frame, the system combines adjacent-frame pixel change and grayscale-vector similarity:

$$s_t = \alpha\left(1 - \frac{1}{255N}\sum_j |I_t^{(j)} - I_{t-1}^{(j)}|\right) + (1-\alpha)\cos(\mathbf{x}_t, \mathbf{x}_{t-1}). \tag{7}$$

A check occurs only when $s_t < \delta$, cooldown has ended, and at least k consecutive frames have changed. This triggers inspection, not necessarily a response. The router selects life, profile, or task memory, and query–memory matching uses

$$score(q_t, m_i) = \lambda\cos(\mathbf{e}_{q_t}, \mathbf{e}_{m_i}) + (1-\lambda)overlap(q_t, m_i), \tag{8}$$

returning Top-K evidence R_t above η. Active tasks and safety facts take priority, followed by confirmed profiles and general life history, with recent and trustworthy records preferred.

4.3 Evidence-Constrained Proactive Decision

Each candidate c_j is checked for scene compatibility $u(c_j, x_t)$ and evidential support $v(c_j, R_t)$. The response model produces a candidate $\hat{a}_t$ and confidence ρ_t, with final decision

$$a_t = \begin{cases} \hat{a}_t, & \rho_t \ge \rho_0 \wedge verify(\hat{a}_t, R_t, \mathcal{I}_t) = 1, \\ \varnothing, & \text{otherwise.} \end{cases} \tag{9}$$

$verify$ checks action timing, historical grounding, profile relevance, task validity, duplication, and risk. A valid dynamic response marks its task *completed*; profile responses enter recent history for cooldown and semantic deduplication. Thus visual change and relevant intent remain distinct from a justified interruption.

5 Experiments

5.1 Experimental Setup

Study 1 uses 65 proactive-response samples from seven days of one participant's recordings, each containing approximately 7–8 valid video hours. Studies 2 and 3 use the 771 thirty-second segments from Day 1 (6:25:30). Study 2 evaluates two six-attribute counterfactual profiles over 1,542 segment–profile decisions; Study 3 contains 12 tasks, 26 `CREATE/MODIFY/CANCEL` events, and 9,252 segment–task pairs. Videos are decoded at 1 FPS without subtitles. Local

Table 1: Study 1 baseline comparison. All metrics except F1 are percentages.

Method	SAA	Recall	Precision	F1
TimeChat-Online	50.00	20.00	23.08	0.2143
Qwen2.5-VL-7B	42.86	**64.62**	21.42	0.3217
Knowing All You Need	**64.00**	38.46	**36.00**	**0.3719**

Table 2: Study 1 component ablation. All metrics except F1 are percentages.

Configuration	SAA	Recall	Precision	F1
Lightweight memory	40.00	20.00	15.38	0.1739
Without evidence gate	42.86	35.38	13.04	0.1906
Complete approach	**64.00**	**38.46**	**36.00**	**0.3719**

Qwen3-VL-2B-Instruct handles scene analysis, routing, and gating; cloud Qwen3-VL-8B-Instruct makes the final `RESPOND`/`SILENCE` decision from up to ten frames (temperature 0.1; confidence threshold 5/10). No annotation or future interaction is exposed at inference.

Evaluation metrics. (i) Basic response: SAA measures candidate-scene judgments. With N_+ required, N_r emitted, and N_c correct responses, $Recall = N_c/N_+$ and $Precision = N_c/N_r$; F1 is their harmonic mean. **(ii) Profile response:** Accuracy, Precision, Recall, and F1 are computed separately for P01 and P02; window recall counts annotated windows with at least one correct response. **(iii) Dynamic tasks:** Segment–task Precision/F1 measure response decisions, window recall measures timely triggering, and state accuracy measures lifecycle transitions. Canceled and repeated responses are also inspected.

5.2 Three Progressive Studies

The studies progressively add life history, stable profiles, and dynamic tasks. They test history grounding, relevant-attribute selection, silence, lifecycle maintenance, timely triggering, cancellation, and one-shot behavior, revealing how errors shift from content selection toward timing and state management.

Study 1: basic proactive response. Table 1 shows that Knowing All You Need improves F1 by 0.0502 over the strongest baseline. Qwen2.5-VL-7B has higher Recall but lower Precision, indicating that more frequent responses are not necessarily more helpful.

Study 1 ablation. As Table 2 shows, lightweight memory lacks sufficient historical evidence. Removing evidence gating increases Recall slightly but sharply reduces Precision: relevant retrieval alone does not make the current moment actionable. Hierarchical memory and gating provide the strongest balance.

Study 2: profile-driven proactive response. Figure 2 reports P01 and P02 separately. P01 reaches two of seven response windows, while P02 reaches one. P01 tends to interpret relevant objects as action points too early; P02 primarily misses responses. Their high Accuracy reflects the many silent segments, so F1 and window recall are more diagnostic. True positives select the correct profile memories, and all emitted responses pass format and direct-helpfulness checks, although the sample is too small to establish stable routing.

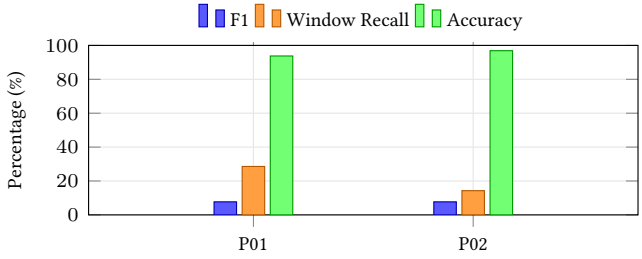


Figure 2: Day 1 user-profile experiment results.

Table 3: Day 1 dynamic-task interaction results (%).

Precision	F1	Window Recall	State Accuracy
25.00	19.35	30.00	76.92

Study 3: dynamic interaction tasks. All events are read, and Table 3 summarizes decision, timing, and state performance. One-shot state eliminates repeated responses, but early triggering completes both canceled tasks before cancellation arrives. The lifecycle mechanism operates, while action-onset localization remains the dominant error because premature responses disrupt later modification and cancellation.

6 Discussion, Conclusion, and Future Work

Knowing All You Need unifies profiles, lifecycle-aware tasks, and life history. Across three studies, one pipeline progresses from history-grounded assistance to stable profiles and mutable tasks. State gating suppresses redundant responses, while evidence connects scenes to earlier events. Yet a relevant response can fail before an action becomes useful or after task state changes, making precise timing a central systems problem.

Evaluation remains limited in participants, days, and authentic profiles; sparse events impair action-onset localization. Future work will expand deployments, add user-confirmed profile updates, and evaluate concurrent tasks. Adaptive sampling, calibrated uncertainty, and personalized interruption thresholds may improve timing. Hybrid execution should be assessed for latency, energy, storage, and privacy-preserving synchronization.

7 User Benefits, Ethical, and Safety Impacts

The assistant helps users recover facts, receive personalized suggestions, and manage reminders. Users should inspect evidence and profile attributes, correct state, pause proactive behavior, and delete memories or tasks.

Risks include surveillance, discrimination, nonconsensual recording [1, 4], fabricated tasks, and harmful advice. Mitigation requires recording indicators, encryption, retention limits, bystander filtering, rate limits, high-risk blocking, and immediate override. Profiles require user confirmation and fairness audits.

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